LEONARDO MEDINA

Forestry Technician | GIS Analyst

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PROFESSIONAL PROFILE

Forestry Engineer with experience in agroforestry systems management, forest inventories, floristic surveys, plant production and environmental analysis. I combine geospatial analysis (GIS), remote sensing, and data analysis to produce maps, manage geospatial databases, technical reports, and applied solutions for land and risk management.

CORE SKILLS

Forestry & environmental: forest inventories, floristic surveys, forest mensuration, plant health, agroforestry, nurseries and greenhouses, botany and taxonomy, conservation, and risk management.
GIS: QGIS, ArcGIS Pro/Online, Google Earth Engine, remote sensing, hydrologic analysis, thematic mapping, and spatial SQL (DuckDB and PostGIS).
Data & automation: Python (Pandas, GeoPandas, Rasterio, Shapely), SQL, Excel, Power BI, ETL, statistical analysis, and Git version control.

PROFESSIONAL EXPERIENCE

Environmental Advisor | Unidad de Respuesta Operativa (URO) - Milla River Early Warning System *(Mérida, Venezuela) | Apr 2026 - Present | Remote*

- Developing the environmental, geospatial, and hydrometeorological components for the modernization of the community-based Milla River flash-flood early warning system.
- Conducting a biophysical, geomorphological, and hydrological characterization of the watershed by integrating topographic, hydrographic, soil, lithologic, land-cover, and critical-infrastructure data.
- Automating the download, cleaning, quality control, and analysis of meteorological and geospatial data with Python.
- Producing thematic maps, technical reports, and reproducible analytical outputs.

Data Analyst | Transportation-Sector Client *| Jun - Dec 2025*

- Designed and implemented ETL workflows with Python and SQL to integrate, clean, and transform data from Excel, CSV, and PostgreSQL sources.
- Performed geospatial analysis of operational data with Pandas, GeoPandas, and mapping tools to identify patterns by station, route, and location.
- Built automated Power BI dashboards to monitor demand, revenue, service incidents, and operational performance.

Data Analytics & Automation | Odontology Coach *| Feb 2025 - Jan 2026*

- Designed and automated clinic operations tracking through more than 15 performance indicators related to patients, treatments, and campaigns.
- Developed data capture, transformation, and validation workflows in Google Sheets using Google Apps Script (JavaScript).
- Produced automated operational reports to streamline performance monitoring and reduce manual work.

Field Forester & Botanical Consultant | Botanical Garden of Mérida - Universidad de Los Andes *| Apr - Aug 2019 | Apr 2021 - Jun 2022*

- Conducted floristic inventories of more than 200 tree, shrub, and herbaceous species, including GPS georeferencing, botanical sample collection, and detailed photographic documentation.
- Performed taxonomic identification using specialized literature, botanical keys, sample comparison, and consultation with experts.
- Reviewed and synthesized specialized sources on species ecology, geographic distribution, and common uses.
- Created thematic maps in QGIS; project results informed three botanical publications: a tree identification key, a digital dendrological catalog, and an article on commonly used plants.

Agroforestry Research Assistant | Instituto de Investigaciones Agropecuarias (IIAP), Universidad de Los Andes *| 2015 - 2019*

- Performed silvicultural management of a cacao-based agroforestry system with timber species including *Cedrela odorata, Swietenia macrophylla, Tabebuia rosea, Cordia thaisiana*, and *Cordia alliodora*, as well as experimental greenhouse crops.
- Monitored system development and plant health through forest and phytosanitary inventories, forest mensuration, and physiological measurements including photosynthesis, fluorescence, and water stress.
- Implemented silvicultural and crop-management practices including pruning, fertilization, phytosanitary treatments, irrigation, weed control, and cacao harvesting.
- Performed statistical analysis of agroforestry field and experimental data.
- Supported technical assistance to local agricultural producers on crop management, nursery production, and fertilization and irrigation planning.

Environmental Consultant | Fundación Programa Andes Tropicales *| Oct - Nov 2021*

- Conducted floristic surveys and identified more than 60 vascular plant species in high-elevation páramo ecosystems within a national park as part of a lake and wetland conservation initiative.
- Analyzed floristic distribution patterns across elevation gradients and in relation to proximity to water bodies.
- Developed a georeferenced dataset, botanical reports, and thematic maps in ArcGIS to support conservation planning.

Field Technician | Latin American Cloud Forest Monitoring Network on Climate Change *| Annual Monitoring Campaigns | 2016–2018*

- Monitored permanent forest plots through species identification and measurement of forest mensuration variables, stem condition, and plant health in accordance with established protocols.
- Validated and processed records in Excel to calculate growth, basal area, mortality, and recruitment.

EDUCATION

Forestry Engineering - Universidad de Los Andes (ULA), Mérida, Venezuela | 2020

CERTIFICATIONS

- Visualizing Land Cover and Land Use Change with Satellite Imagery - NASA ARSET | 2026
- Google Data Analytics - Google | 2025
- Data Analysis with Python - IBM | 2025
- Renewable Natural Resources and Environmental Planning - CIDIAT-ULA | 2024

PUBLICATIONS

- Soto, C., Medina Linares, J. L., & Gámez, L. E. (2024). Catálogo dendrológico del Jardín Botánico de Mérida, estado Mérida, Venezuela. Universidad de Los Andes. ISBN 978-980-11-2149-7.
- Soto, C., & Medina Linares, J. L. (2023). Plantas de utilidad común del Jardín Botánico de Mérida. Revista de la Facultad de Farmacia. DOI: 10.53766/REFA/2024.65.2.03.
- Medina Linares, J. L., Soto, C. A., & Gámez, L. E. (2021). Clave de identificación de árboles del Jardín Botánico de Mérida por medio de caracteres vegetativos. Pittieria, 44, 104-127.

LANGUAGES

Spanish: Native | **English:** C1 - Professional working proficiency